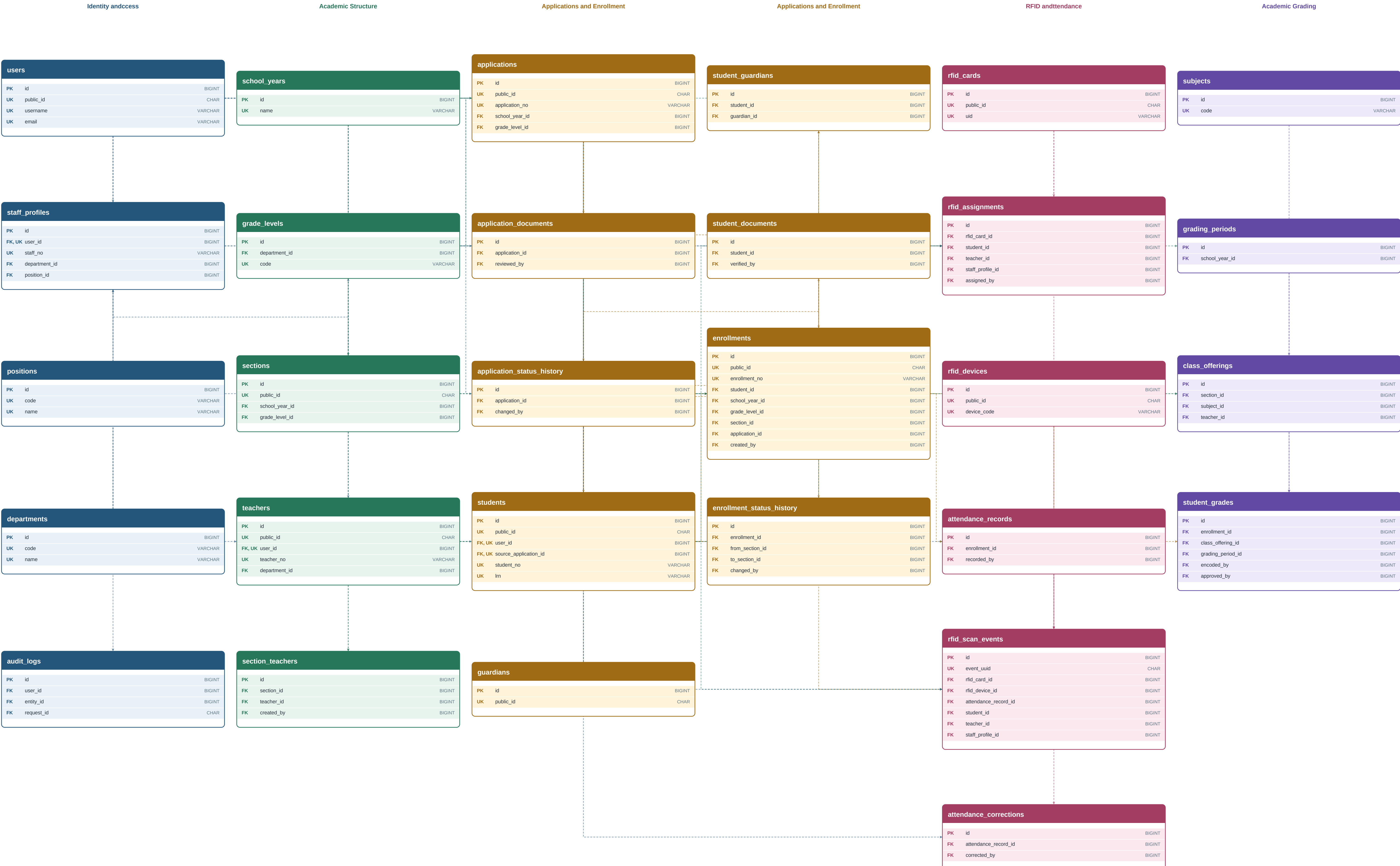


School Management System - Full Relationship Mapping

All entities and key fields; arrows point from parent to dependent entity



Cardinality, probable foreign key, and implementation meaning

Parent	Cardinality	Child	FK in child	Meaning
users	1 to 0..1	staff_profiles	user_id, user_id	has staff profile
users	1 to 0..1	teachers	user_id, user_id	may authenticate
users	1 to 0..1	students	user_id, user_id	may authenticate
users	1 to many	audit_logs	user_id, user_id	performs
users	1 to many	application_documents	reviewed_by	reviews
users	1 to many	application_status_history	changed_by	changes
users	1 to many	enrollments	created_by	creates
users	1 to many	attendance_records	recorded_by	records
users	1 to many	attendance_corrections	corrected_by	corrects
users	1 to many	rfid_assignments	assigned_by	assigns
positions	1 to many	staff_profiles	position_id	classifies
departments	1 to many	staff_profiles	department_id	contains
departments	1 to many	teachers	department_id	contains
departments	1 to many	grade_levels	department_id	groups
school_years	1 to many	sections	school_year_id	contains
school_years	1 to many	applications	school_year_id	receives
school_years	1 to many	enrollments	school_year_id	contains
school_years	1 to many	grading_periods	school_year_id	divides
grade_levels	1 to many	sections	grade_level_id	offers
grade_levels	1 to many	applications	grade_level_id	requested_by
grade_levels	1 to many	enrollments	grade_level_id	places
sections	1 to many	section_teachers	section_id, section_id	has
teachers	1 to many	section_teachers	teacher_id	assigned_to
applications	1 to many	application_documents	application_id, application_id	includes
applications	1 to many	application_status_history	application_id, application_id	tracks
applications	1 to 0..1	students	source_application_id	becomes
applications	1 to 0..1	enrollments	application_id, application_id	produces
students	1 to many	student_guardians	student_id	has
guardians	1 to many	student_guardians	guardian_id	supports
students	1 to many	student_documents	student_id	owns
students	1 to many	enrollments	student_id	enrolls
sections	1 to many	enrollments	section_id, section_id	contains
enrollments	1 to many	enrollment_status_history	enrollment_id	tracks
rfid_cards	1 to many	rfid_assignments	rfid_card_id	assigned_through
students	1 to many	rfid_assignments	student_id	may_own
teachers	1 to many	rfid_assignments	teacher_id	may_own
staff_profiles	1 to many	rfid_assignments	staff_profile_id	may_own
rfid_cards	1 to many	rfid_scan_events	rfid_card_id	scanned
rfid_devices	1 to many	rfid_scan_events	rfid_device_id	captures
students	1 to many	rfid_scan_events	student_id	resolved_as
teachers	1 to many	rfid_scan_events	teacher_id	resolved_as
staff_profiles	1 to many	rfid_scan_events	staff_profile_id	resolved_as
enrollments	1 to many	attendance_records	enrollment_id	receives
attendance_records	1 to many	rfid_scan_events	attendance_record_id	supported_by
attendance_records	1 to many	attendance_corrections	attendance_record_id	corrected_by
sections	1 to many	class_offerings	section_id, section_id	schedules
subjects	1 to many	class_offerings	subject_id	offered_as
teachers	1 to many	class_offerings	teacher_id	teaches
enrollments	1 to many	student_grades	enrollment_id	receives
class_offerings	1 to many	student_grades	class_offering_id	generates
grading_periods	1 to many	student_grades	grading_period_id	groups

Delete-behavior guidance

RESTRICT: students, enrollments, sections, attendance, grades, and other official history. SET NULL: optional actor and optional parent references.

CASCADE: true dependent children such as application documents/status history and junction rows. Prefer deactivation/soft deletion for operational master records.